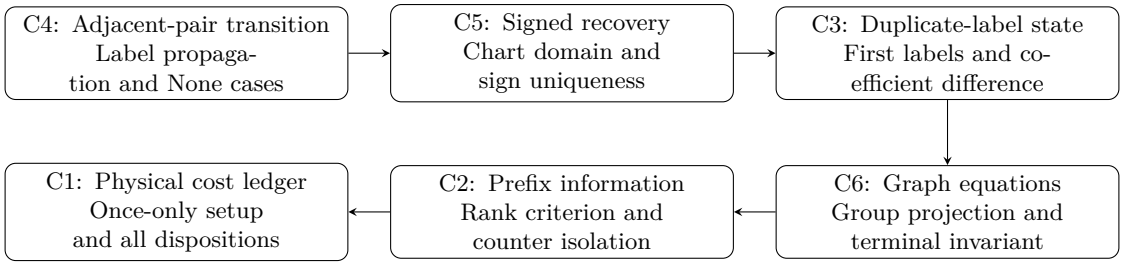
Relation information and Kummer source contracts

Source-only audit, 1 November 2026

The audit identifies six composition contracts joining physical execution, signed labelled divisors, graph relations, target information and complete charged work. Existing implementations and review artifacts are reused. No experiments, new target trajectories, cloud actions, timing measurements or theorem certifications were performed. **Scope preserved:** 1,260 targets. Complete cost ratio and natural yield remain null. This packet is a source-contract map, not a completion certificate.



Arrows indicate required composition; a node does not assert that its obligation is proved. The editable SVG contains the same dependency sequence. **Historical saved observations.** The prior portfolio inventory contained 140 files, all matching their recorded hashes during the preceding read-only audit. Its standardized table records five factor-base arms with 1,260 rows each, two law arms with 1,260 each and a Kummer differential arm with 1,260. Historical pencil coverage records 543 verified candidate keys, one capped-only key and 716 without indexed candidate rows. The later C1 gate records 7/18 observations, 4/15 measured, no complete pairs and two timeouts. These are saved-artifact counts, not new runs or scalar replays. The retained duplicate archive analysis reports 420 dispositions: 218 ineligible and 202 compared, comprising 174 first observations and 28 redundant observations. The actual source uses retained coordinate witnesses, so these counts do not estimate naturally generated duplicate yield.

C1-physical-cost-partition

Existing evidence. archive_target emits setup_checks_field and constructor_field repeatedly for seven retained cases. Its explicit metadata says these copies are nonadditive and aggregate once per physical archive_row. Analysis already groups setup by physical row and reports no consistency errors. **Unproved or unbound transition.** No complete timing or cost denominator. Constructors contain Ring primality and allocation work outside field counters; input parsing, curve construction, hashing, memory, comparisons and square-root nonresidue search also need their own accounting. This is retained-coordinate input, not a naturally generated production stream. **Concrete contract.** For immutable input archive hash H, let P=(H,archive_row). Require exactly one setup and constructor record per P, identical repeated metadata, and one disposition per (P,case_index). Total charged work is sum_P setup(P)+constructor(P) plus all disjoint per-case intervals and declared run-global work. Never sum copied setup on each case; never equate target_index with physical identity across archives. Preserve failure, skipped, poisoned and post-terminal dispositions with explicit charged/non-production reason. **Next source/proof work.** Write schema-level bijection and disjoint-interval checker contract against existing ledger, not a replacement aggregate. **Source references:** archive:855-881, archive_entry:2, accounting.

C2-production-prefix-information

Existing evidence. information computes rank(A), rank([A b]), rank([A b c]); consistent iff full rank equals augmented-target rank; target determined iff consistent and rank gain one. Adapter validates then normalizes and flips only final RHS column. Independent sign review exists. **Unproved or unbound transition.** The diagnostic performs six rank computations through two information calls and mutates thread-local arithmetic counters. It has no production prefix completeness bridge, emitted-stream custody or policy experiment. **Concrete contract.** For each production row identity, bind curve/target/ring, physical source event, accepted signed factors, source labels and exact augmented sign convention. Prefix export must be bijective with accepted preterminal events and preserve rejected/duplicate dispositions separately. Compute offline diagnostic after snapshotting production counters; either isolate diagnostic counters or account them as a distinct analysis phase. At each prefix target information is consistency AND rank([A b])-rank(A)=1, not total rank or cycle count. **Next source/proof work.** Specify adapter for actual retained emitted prefixes and count boundary; reuse existing rank and sign routines. **Source references:** information:34-56, adapter:1-26, adapter_review.

C3-duplicate-label-invariant

Existing evidence. Signed coefficient-difference lemma and independent 12-case review exist. Native Labels stores first sign-normalized pair and suppresses duplicate graph edges. Candidate is verified against Q. **Unproved or unbound transition.** Labels::ingest accepts D without checking validity or D=aG+bQ. Constructor checks rG=rQ=0 but not Q in <G>. Retained harness supplies the stronger equality with planted-scalar setup; prospective production must establish it from initialized and propagated labels. **Concrete contract.** In fixed target context with ord(G)=prime r, Q=kG and D=aG+bQ, canonical key/sign must satisfy sigma D=C and both labels normalize by sigma. Preserve first pair unchanged. Equal keys imply delta_a+delta_b*k=0; db=0 forces da=0, else verified candidate is -da/db. Closure theorem must connect every actual transition to the label invariant; rQ=0 alone is not the subgroup premise. **Next source/proof work.** Formalize native table-state preservation and caller-to-ingest label transport; do not redo the existing elementary duplicate lemma as if new. **Source references:** duplicate:1-22, archive:863-879, lemma, lemma_review.

C4-original-G-transition-closure

Existing evidence. advance implements geometric step, optional validation, then signed fallback only if geometric step returns None. Original-G fixture has 63 transitions per arm and candidate63 exact recoveries. **Unproved or unbound transition.** When validate_transition=false and geometric step returns Some, no internal full signed validation executes. Finite fixture excludes successful repair and Delta-zero branches. Signed fallback can return None after validity/projection failure; thus universal total closure is not proved. **Concrete contract.** Define Adj(k,kp,a,b): both nonzero valid Kummer representatives of D=aG+bQ and D+G, with canonical labels and rG=rQ=0. Under explicit projection/chart/repair domain, advance Some must preserve Adj at (2a+choice mod r,2b mod r); choice in {0,1}. Separately characterize every None and prove retry policy preserves target identity and charges abandoned work. Word arithmetic requires bound ensuring 2a+choice and 2b do not overflow ($r \leq 2^{\wedge}63$ suffices for a,b<r). **Next source/proof work.** Prove source-return soundness branch by branch before claiming total closure; enumerate chart-domain predicates as preconditions rather than assuming fallback always succeeds. **Source references:** transition:235-324, original_status.

C5-signed-lift-uniqueness

Existing evidence. lift dispatches general recovery, D_zero exceptional search, Delta_zero lift and degree-one boundary lift. Boundary implementation filters candidate validity and two projective equalities and returns only a unique candidate. **Unproved or unbound transition.** Kernel connection from projective equality to unique signed Jacobian value is not supplied by fixture counts. Ambiguous torsion anchors, invalid/singular charts and no-match outputs remain explicit cases. **Concrete contract.** Given valid anchor G and neighbor pair (kappa(D),kappa(D+G)), prove each Some result equals D under the exact chart and sign-uniqueness assumptions; prove zero/ambiguous/no-match outputs are classified without treating projection equality alone as sign equality. General and exceptional implementations need separate source bindings; preserve actual nonsquare/singular failures. **Next source/proof work.** Bind existing boundary and original-G reviews to precise sufficient uniqueness premises; prove one shared result interface across recovery branches. **Source references:** transition:175-219, original_status.

C6-graph-to-group-and-terminal

Existing evidence. Graph adapter factors recovered split degree-two D, assigns canonical signs and checks factor sum in fixtures. Independent dense elimination checked emitted prefixes; synthetic terminal controls exist. Graph review explicitly reports zero geometric terminal returns. **Unproved or unbound transition.** A compatible F_r scalar coordinate for every factor-base point class is not established by ord(G)=r. Geometric graph terminal/duplicate/repeated-root branches were not exercised in that fixture. Verification-on-return is not a proof of target extraction completeness. **Concrete contract.** Supply an actual homomorphism phi from the subgroup generated by all relevant divisor classes to F_r with phi(G)=1 and phi(Q)=k, or restrict points to a justified subgroup/projection. Then prove every inserted su*Xu+sv*Xv=a+b*k row follows from D=aG+bQ and factor decomposition. Prove signed union-find potential invariant and that a nonzero eliminated target coefficient yields k; preserve zero-information cycles and consistency failures. **Next source/proof work.** Identify the precise group-model/projection premise from existing construction and state the graph invariant against it; do not infer it from r-torsion tests. **Source references:** graph:10-16, graph_review, lemma.

Accounting partition

All intervals reflect the fixed retained harness. Production-vs-reference split needs frozen protocol; all actual incurred costs remain preserved. **Physical identity:** (input_archive_sha256, physical_archive_row, case_index) **setup_once**

- setup_checks_field
- constructor_field

per_case_disjoint

- label_reference_field
- eligibility_field
- key_reference_field
- adapter.field
- adapter scalar_after minus scalar_before
- graph.processing_field
- graph factor_verification_field
- graph ring_after minus ring_before
- terminal_reference_field

nonadditive_or_external

- setup copies repeated on each case
- graph state snapshots
- walk/recovery custom counters may overlap field counts; require mapping before sums
- dense reference work and offline rank work
- host/runtime elapsed and memory not inferred from field counters

Source custody

Exact absolute paths and SHA256 hashes are preserved in contracts.json. The following source keys bind the report statements. **archive**: `followup-continuation-v2/packets/retained-duplicate-executed/src/pruned.rs`
`e77db9e741da1d74b3dad97bf8f52c1128d4ac9e61eb0ea8dd62ddc923c21fcf`

archive_entry: `followup-continuation-v2/packets/retained-duplicate-executed/src/check.rs`
`1f685079ab2ec640c271538db81b4a1d4a08fa7784424eeec07ba3efee636385`

duplicate: `followup-continuation-v2/packets/retained-duplicate-executed/src/duplicate.rs`
`33e74627a4b513b932e10b65f52fb5952ff8bc0473b3e0e76366689dc2494fd9`

accounting: `followup-continuation-v2/packets/retained-duplicate-executed/analysis.json`
`7024ed785cade871769b90a8d983a7e1828ccc59289c0f1225091a7a98f33fef`

graph: `followup-continuation-v2/packets/kummer-graph-integration/src/graph.rs`
`a612c5f4f2bea7c2573d147e4e54bfd2775fcf4057d2b11dd3ccf9cc63c58093`

transition: followup-continuation-v2/packets/kummer-original-gstep/src/pruned.rs
149d55205b5edd872610350a2ddb7bf872d248f0e46f131b0091d618d33c6e5e

lemma: followup-continuation-v2/packets/kummer-target-rank/LEMMA.md
2a0d9c3a61941e6a066213e32890a0f5416098cc5b97f13dcbe0d5e23be83a20

lemma_review: followup-continuation-v2/packets/kummer-target-rank-review/status.json
38eaf142ca7e732b7759c47d28b6f19be783a61c2e24bc64b488243b9e01955b

graph_review: followup-continuation-v2/packets/kummer-graph-review/REVIEW.md
4830a8ec0a321a070fbddc5adb20cde532d824d5417643edc9bb54f9d261a236

adapter: followup-continuation-v2/relation-adapter/src/lib.rs
5c576f9675df206f463dee8169b860876b98c618f51b406323a35c7ed8423120

adapter_review: followup-continuation-v2/packets/relation-adapter-review/REVIEW.md
d647ec25ae718a2c87dcbebf0b471162cef146b6a4f7bf4fcac7261484af70bf

information: five-followups/rank-information/src/lib.rs
602b0ebc368df64b3fdc019c3d2fed05ce3f12d437c8d297d4a8af3a37911f68

original_status: followup-continuation-v2/packets/kummer-original-gstep/status.json
9c209baec29d03c45855f24fdb305ad32a9f236af213483c7f9916abeca29b70

duplicate_status: followup-continuation-v2/packets/kummer-duplicate-adapter/status.json
9d18f55aec8d72f817da624cdd3d9ba6e7e578380eceae5d02eefe9d4f439d38